

Q What do you remember about the geometry of linear algebra?

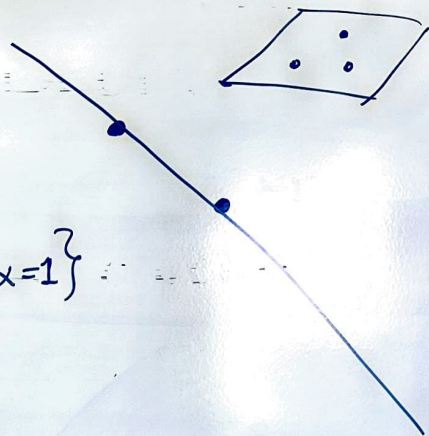
- determinant $\longleftrightarrow$ signed area
- normed vector spaces have distances.
- Orthogonality $\begin{matrix} \nearrow a \cdot b = 0 \\ \searrow \perp \end{matrix}$
- dot product $\longleftrightarrow$ length in a direction
- rank $\longleftrightarrow$ dimension
- projections, rotations
+ general (often linear) transformations
- angle b/w vectors (using $\cdot$)
- affine spaces

Affine Hull

$P \subset \mathbb{R}^d$ a point cloud

$$\text{aff}(P) = \left\{ \sum_{p \in P} \alpha_p p, \text{ such that } \sum \alpha = 1 \right\}$$

note: an affine space is like a vector space, just with no "origin"



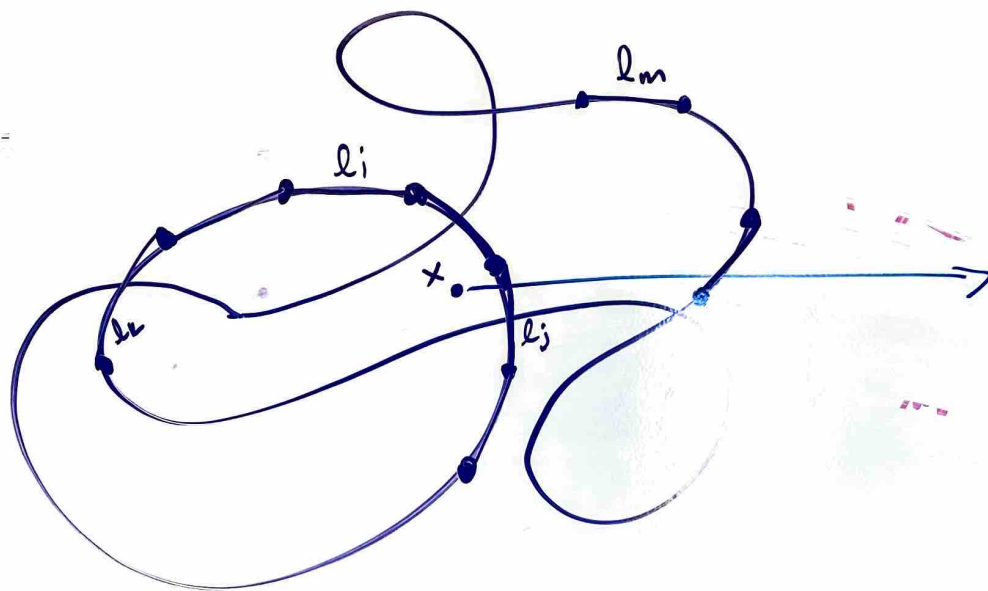
Convex Hull

$P \subset \mathbb{R}^d$ finite

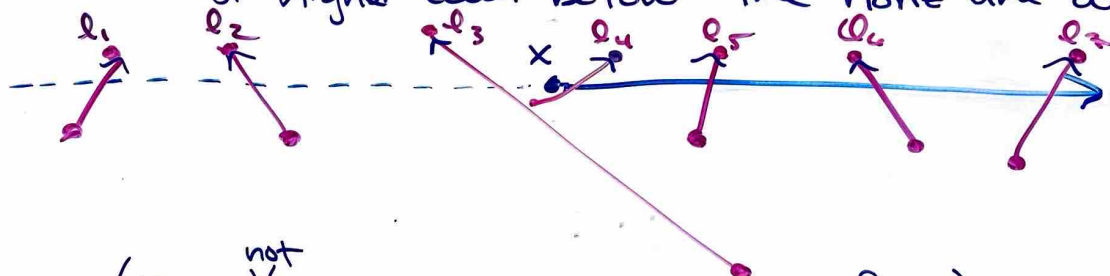
$$\text{CH}(P) = \left\{ \sum_{p \in P} \alpha_p p, \text{ such that } \sum \alpha_p = 1 \text{ and } \alpha_p \geq 0 \forall p \in P \right\}$$

$$\Leftrightarrow \text{CH}(P) = \bigcap_{\alpha} H_{\alpha}, \text{ such that } H_{\alpha} \text{ is a half-space containing } P$$





#1: eliminate line segments w/ lower coord above or higher coord. below the horiz line at x .



(if x is ^{not} "in between" both in $x+y$, know for sure)

#2: orient the line from bottom to top

Ask: is x on the left or the right of that line?

(+)

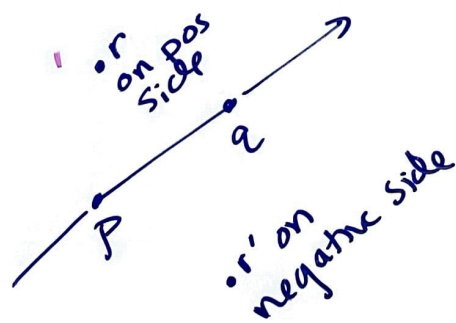
(-)

Right hand rule

- wrist at 1st point
- point pinky finger on 2nd pt
- hand curls towards 3rd point
- when positively oriented.

Jack's Favorite Determinant

$$\begin{vmatrix} 1 & P_x & P_y \\ 1 & q_x & q_y \\ 1 & r_x & r_y \end{vmatrix}$$



line.

Signed Homogeneous Coordinates (in $\mathbb{R}^3$)

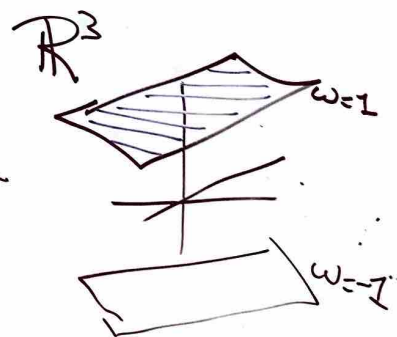
$$p, q \in \mathbb{R}^3$$
$$p \equiv q \text{ iff } \exists \alpha > 0 \text{ s.t. } p = \alpha q$$

(aka: any point emanating from
same ray from origin is the same)

- in math: S^2

- in graphics: $p = (w, x, y)$
 w is always 1 $\uparrow$ homogeneous coordinate

- $w=0$ are pts at infinity



How to work in
oriented projective geometry.

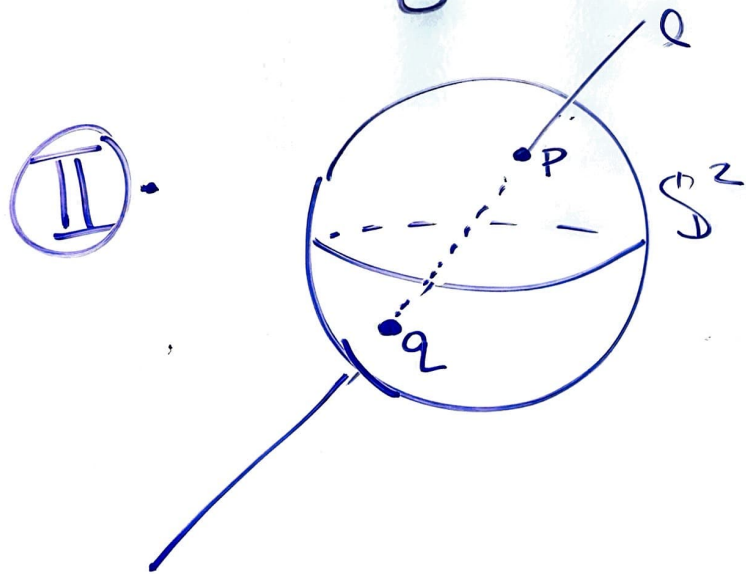
- 2 lines always intersect
- less corner cases more generally

$w=1$

$w=-1$

Projective Plane : 2 models

(I) the set of lines through the origin in $\mathbb{R}^3$



$\left\{ x \in S^2 \mid \begin{array}{l} x \equiv y \text{ iff} \\ x + y \text{ are antipodal} \end{array} \right\}$

$\uparrow$

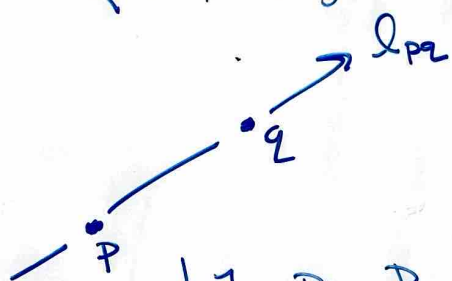
points in S^2 under
an equiv. relation

It is to work in

$(w, x, y) \leftarrow \text{a point}$

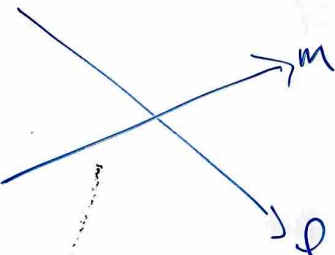
$\langle w, x, y \rangle \leftarrow \text{a line (normal vector)}$

"equation" of
a line through two points

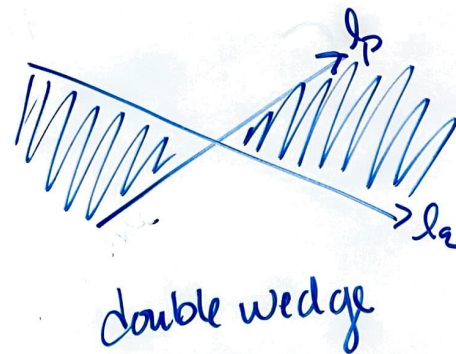
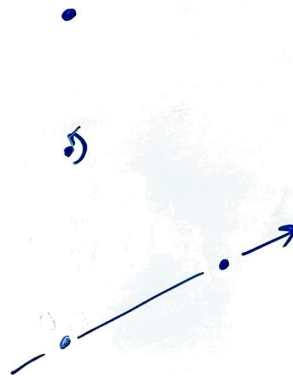
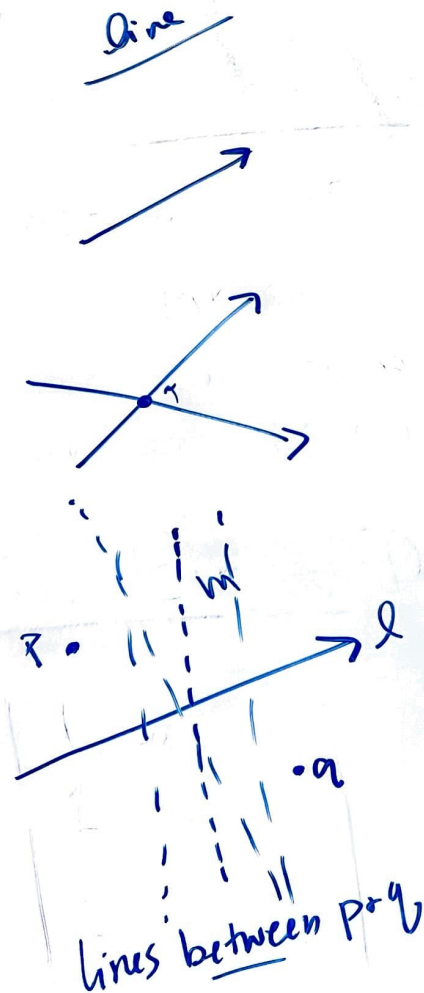


$$l_{PQ} = \begin{vmatrix} 1 & P_x & P_y \\ 1 & Q_x & Q_y \\ \underline{w} & \underline{x} & \underline{y} \end{vmatrix} = \left\langle \begin{vmatrix} P_x & P_y \\ Q_x & Q_y \end{vmatrix}, -(Q_y - P_y), P_x - Q_x \right\rangle$$

$$l_{lm} = \begin{vmatrix} l_w & l_x & l_y \\ m_w & m_x & m_y \\ \underline{w} & \underline{x} & \underline{y} \end{vmatrix}$$



Duality



double wedge

Defn: Knots

A knot K is an embedding of S^1 in $\mathbb{R}^3$.

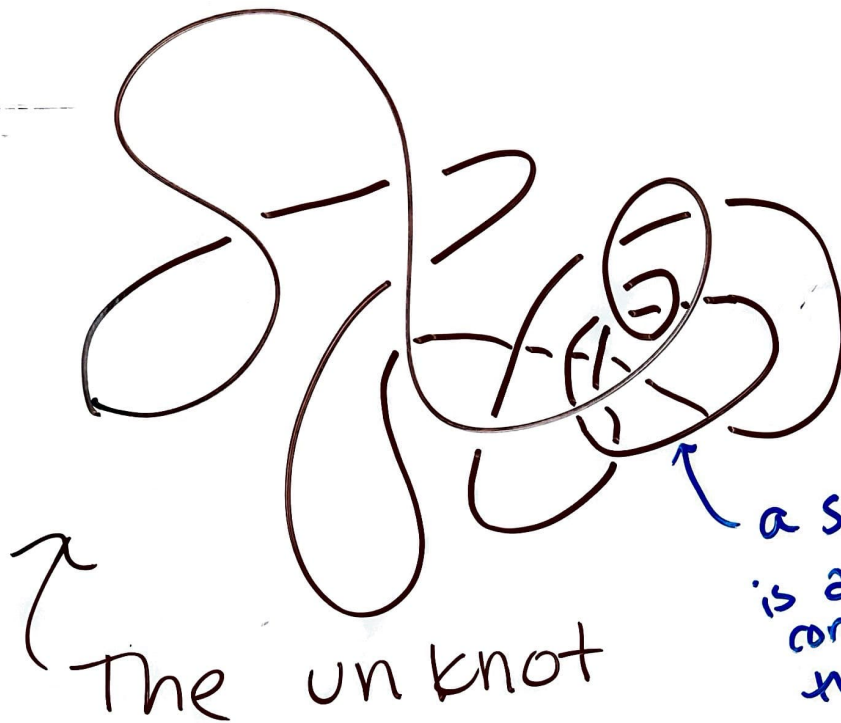
(note: all knots are the unknot in $\mathbb{R}^4$)

A link is $2+$ knots.

A knot/link ^{K/L} drawing is a "nice" projection of K/L into $\mathbb{R}^2$ such that

(i) we have an immersion

(ii) we note under/over crossings



bar trick'. start drawing &
always go "under". you
wind up w/ the unknot.

Reidemeister Moves

II.



OR



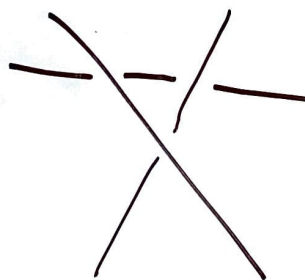
I.



OR



III.



(3rd strand
can be above too)

Hope:

Can always "walk" from
one drawing to another of
the same Knot using these moves.

Yes! we can

Dream:

can do it using no "complicating"
moves

→ NOPE

The Culprit

must use complicating
moves to "unknot" to the
unknot.

Three Colorability

A knot is 3 colorable if
 $\exists$ a drawing such that each
strand is colored one of R, G, B
and at least 2 colors are used

and each intersection sees either
3 colors or 1 color.

$\square$ R-moves preserve this,